

Non-Newtonian Viscosity

Final Lab Report

Unit Operations Lab 1

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Team 3

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Executive Summary:

The purpose of this lab was to determine the apparent viscosity of a 0.8 wt% solution of CARBOPOL in water. Tests were conducted by moving the viscous fluid through capillary tubes for set intervals of 30 seconds. The sample collected after 30 seconds was then weighed and recorded. Tests were ran at 30psi, 40psi, and 50psi using capillary tubes with varying lengths of 0.0762 meters, 0.1524 meters, and 0.2286 meters and varying diameters of 0.001778 meters, 0.001016 meters, 0.013462 meters, 0.0015748 meters. The Ostwald-deWaele viscosity model was used to determine the 'n' and 'm' values of the viscous fluid. For the capillary length of 0.0762 meters and a pressure of 30 psi, $n = 0.39$ and $m = 8.042589463$. For the capillary length of 0.1524 meters and a pressure of 40 psi, $n = 0.2701$ and $m = 8766.704465$. For the capillary length of 0.2286 meters and a pressure of 50 psi, $n = 0.2606$ and $m = 3239.135292$. A graphical representation of apparent viscosity vs. the length to diameter ratio at all three pressures showed that there is a general trend for this non-Newtonian viscous fluid. It was discovered that between a length to diameter ratio of 0 and 20, there is an upward trend of apparent viscosity. Between a length to diameter ratio of 20 and 40, there is a downward trend of apparent viscosity. After a length to diameter ratio of 40, the apparent viscosity remains close to 0.

Introduction

To estimate the transport behavior of the fluids, the viscosity is studied. Viscosity has high dependence on the Reynolds number of the liquid. It is defined as a measure of the fluid's resistance to flow [3]. For Newtonian fluids, the shear stress is directly related to the velocity gradient. For non-Newtonian fluids, this linear relationship is not observed [1].

In this lab experiment, 0.8 wt. % Non-Newtonian Carbopol solution is prepared to study the apparent viscosity. To achieve that, we use the capillary system approach to study the effects of pressure gradient at different flow rates by varying the size of the capillaries. Capillaries of length, 3, 6, and 9 inches with varying diameters were utilized to determine the volumetric flowrates of the Carbopol solution. 27 measurements were made by varying the pressure at the values of 30 psig, 40 psig, and 50 psig. Also, the constants, m and n, were determined using graphs obtained to show the correlation between pressure gradient, volumetric flow, and use the model (Ostwald-de-Waele) utilized to determine the non-Newtonian Fluid's viscosity.

In Ostwald-de-Waele equation is utilized where m and n parameters are determined by plotting the ln-ln graphs between pressure drop and flowrate relations.

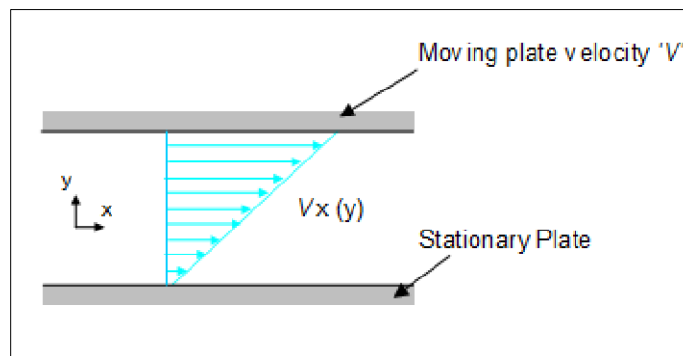
Theory

Taking in consideration a system of two horizontal plates with a fluid in between when one plate is slide at constant velocity, a velocity profile is observed in the fluid. The force necessary to ensure the constant motion of the plate can be determined experimentally. The force is proportional to the velocity gradient in the fluid i.e.

$$\tau_{yx} = -\mu \frac{dv_x}{dy}$$

where,

- the τ_{yx} is the shear stress in x-direction with respect to a normal in y-direction; units: force per unit area
- $\frac{dv_x}{dy}$ is the gradient in y direction of fluid's velocity in x direction; units: 1/s
- μ is the viscosity in fluid; units: Pa*s



Fluid contained in between two parallel plates

In this lab, the viscosity of the fluid is measured by recording and measuring the pressure drop and the flow rates of the non-newtonian fluid through different capillaries. The relation of the fluid's viscosity to the pressure drop and flow rates are given by the Hagen-Poiseuille equation, i.e.

$$Q = \frac{\Delta P \pi R^4}{8\mu L}$$

where,

- Q is the fluid volumetric flow rate; units: m^3/s
- ΔP is the pressure drop across the capillary tube; units: Pa
- R and L are the radius and length of the chosen capillary; units: m
- μ is the viscosity of the fluid; units: Pa*s

The graph of fluid volumetric flow rate (Q) vs. the pressure drop across the capillary tubes (ΔP) will give a line of slope $\frac{\pi R^4}{8\mu L}$ which gives the viscosity of the fluid.

Some  Newtonian fluids can follow an analytical equation of flow rate in tube by the viscosity model of Ostwald-deWaele;

$$m \left[\frac{3n + 1}{n} \right]^n * \left(\frac{Q}{\pi R^3} \right)^n = \left(\frac{\Delta P}{2L} R \right)$$

where, m and n are the parameters of Ostwald-deWaele model. The plot a $\ln \left(\Delta P * \frac{R}{2L} \right)$ vs $\ln(Q/\pi R^3)$ gives parameters of the model. When $n = 1$, then m becomes equal to μ and the equation turns to Hagen-Poiseuille law for a Newtonian pipe fluid.

Results

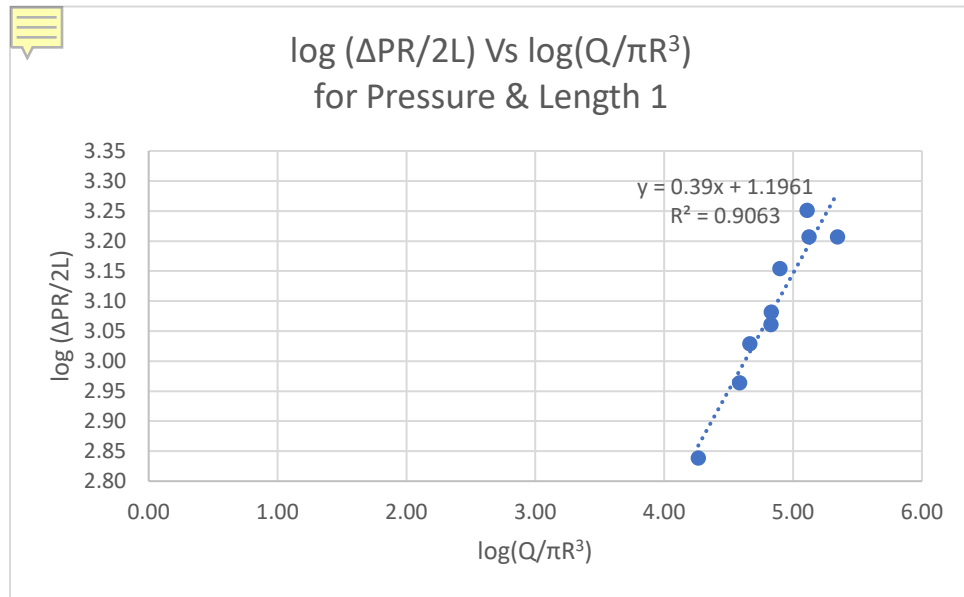


Figure 1: The relation of pressure drop vs flowrate for 3in pipe length and 30 psig

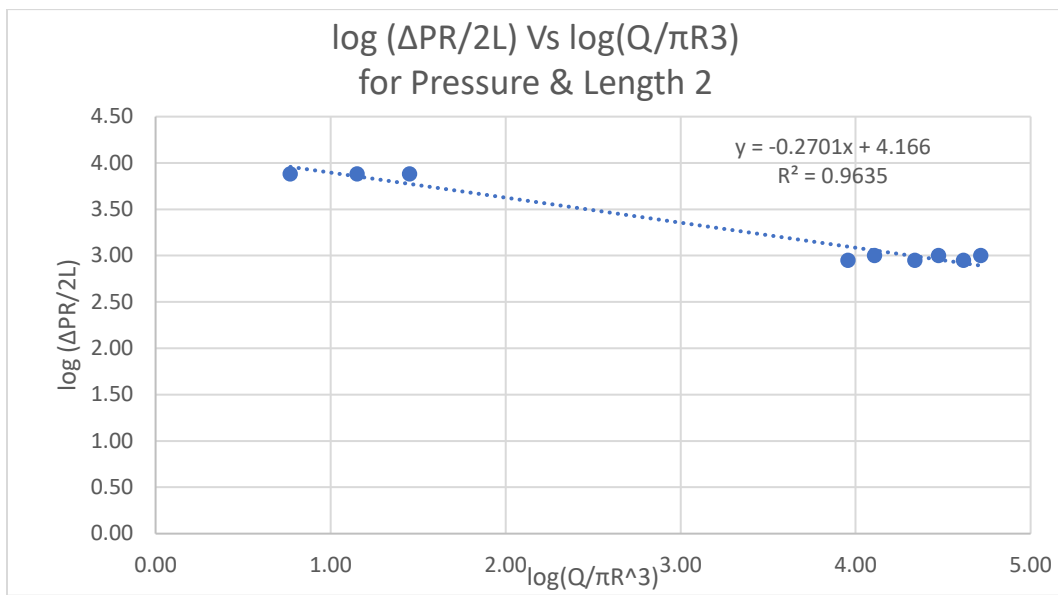


Figure 2: The relation of pressure drop vs flowrate for 6in pipe length and 40 psig

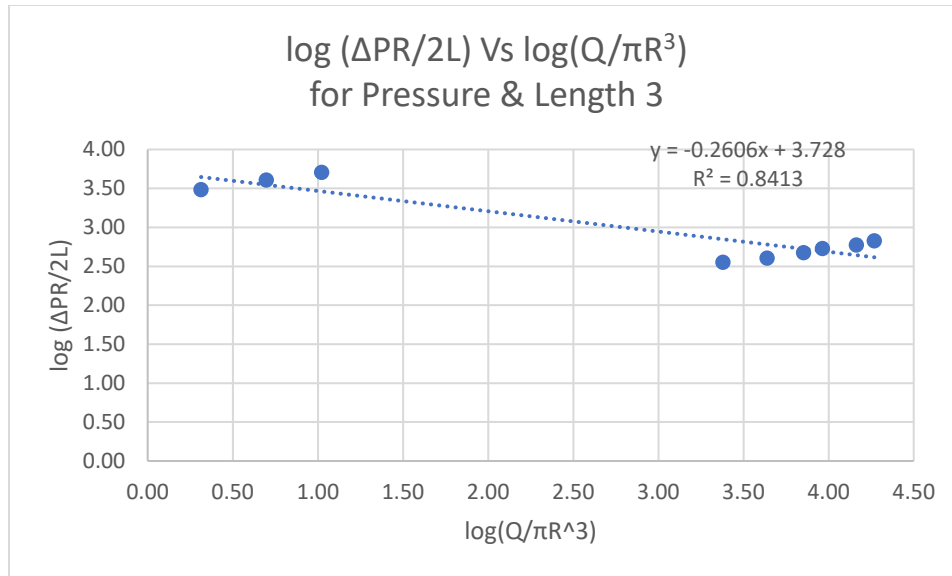


Figure 3: The relation of pressure drop vs flowrate for 9in pipe length and 50 psig

In this experiment, flowrate and pressure drop were used to determine the apparent viscosity. CARBOPOL, the non-Newtonian fluid, obeys the Ostwald-de'Waele viscosity model.

$$\tau_{yx} = -m \left| \frac{dv_x}{dy} \right|^{n-1} \frac{dv_x}{dy} = -\mu_{app} \frac{dv_x}{dy}$$

For fluids that obey the Ostwald-deWaele viscosity model, the following equation can be obtained:

$$m \left[\frac{3n+1}{n} \right]^n * \left(\frac{Q}{\pi R^3} \right)^n = \left(\frac{\Delta P}{2L} R \right)$$

The three graphs above show the general trends of log(dPR/2L) vs. log(Q/R³). The 'n' value can be equated to the slope of the line, while the 'm' value can be calculated from the intercept value. The first graph had values of n = 0.39 and m = 8.042589463. The second graph had values of n = 0.2701 and m = 8766.704465. The third graph had values of n = 0.2606 and m = 3239.135292. The 'n' value is dimensionless while the 'm' value has a unit of Pa*s. The trends of graphs two and three are like each other, while the trend of graph one differs quite a bit from the other two. Graph one differs in the trend line but the high R² values assure the best fit line to each graph.

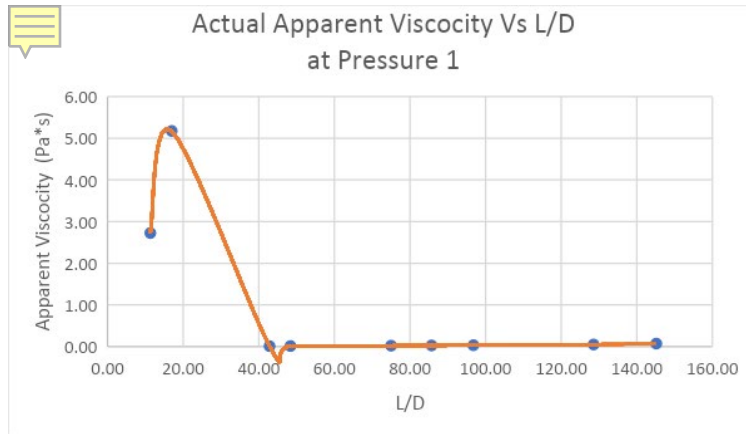


Figure 4: Actual apparent viscosity vs L/D @ 30 psig

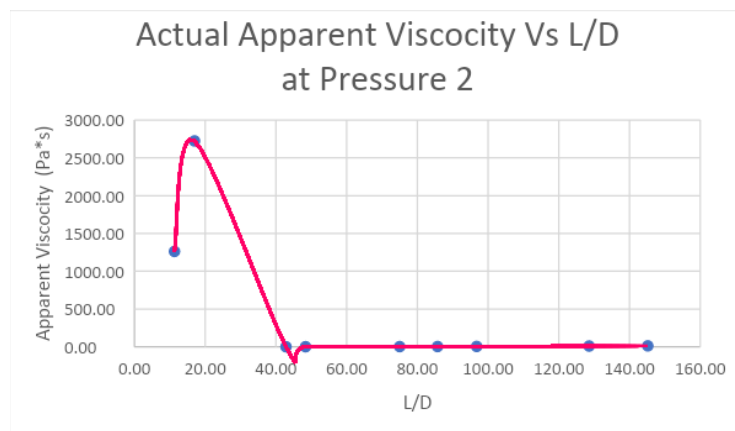


Figure 5: Actual apparent viscosity vs L/D @ 40 psig

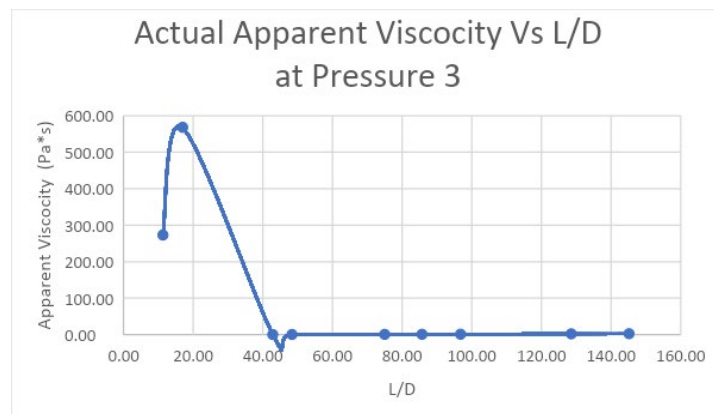


Figure 6: Actual apparent viscosity vs L/D @ 50 psig

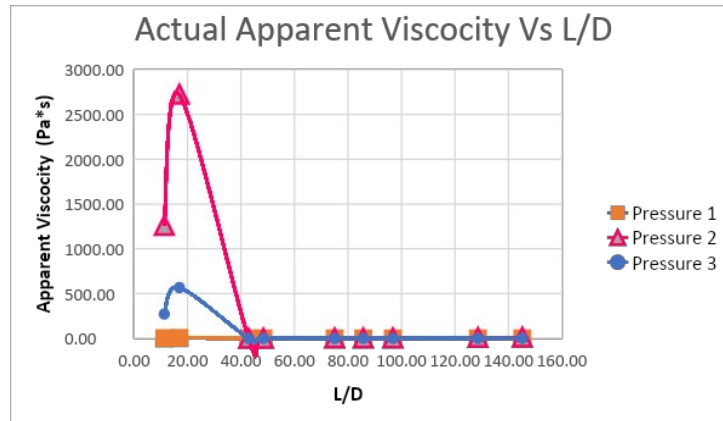


Figure 7: Actual apparent viscosity vs L/D @ combined pressures

The four graphs above show the relation between the apparent viscosity and the length to diameter ratios for the three different pressures. The fourth graph contained the combined data from the first three graphs. The graphical results show that the apparent viscosity vs. the length to diameter ratio lines for each pressure follow a similar trend of increasing, then decreasing. In the fourth graph, it is shown that a maximum apparent viscosity was achieved using a pressure of 40 psi as opposed to 30 psi or 50 psi.

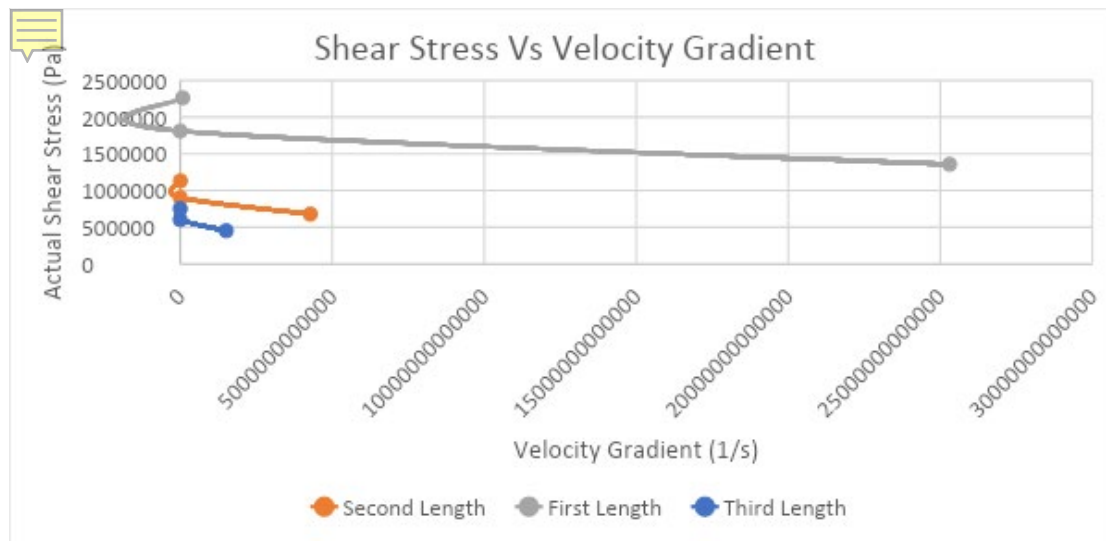


Figure 8: Shear stress versus velocity gradient at 3 in, 6 in and 9 in capillary tube

The graph above represents the relationship between the actual shear stress and the velocity gradient for each one of the three lengths of capillary tube used.

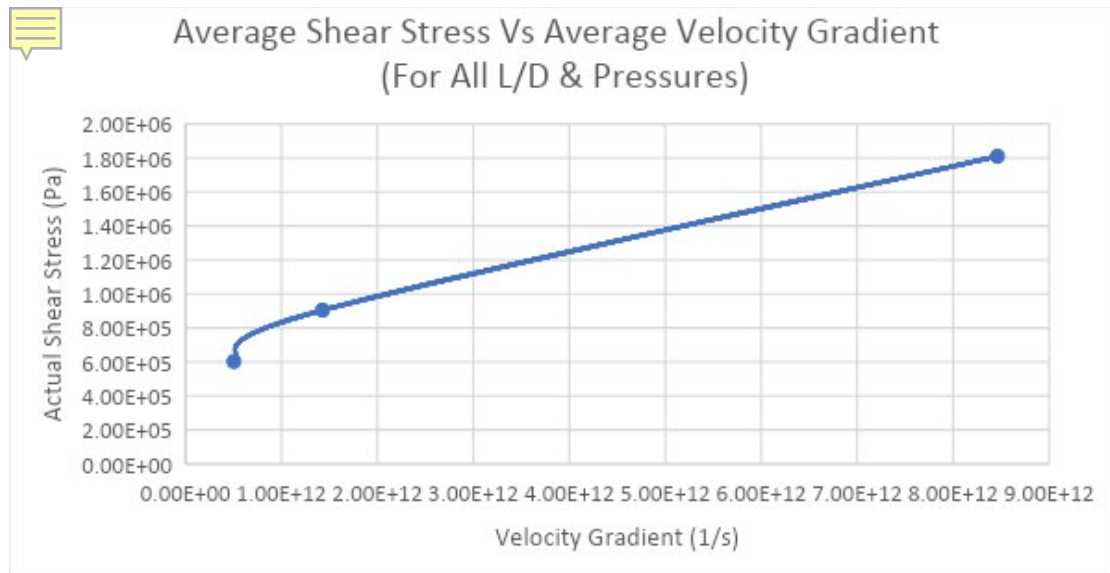


Figure 9: Average shear stress versus average velocity gradient for all L/D and Pressures

The graph above represents the relationship between average shear stress and the average velocity gradient for all the pressure and length to diameter ratios. It appears to be mostly linear with an upwards trend.

Discussion:

The non-Newtonian fluid used for this lab was Carbopol and so it was predicted to have increasing viscosity as the pressure is increased. It was hence observed that as the pressure was increased in the viscometer, the force increased as well and hence making the fluid flowing faster through the only exit point of the vessel i.e. the capillary tubes. It is noted on figures 4-7 that the viscosity did increase in low values with pressure.

The relation between the Length/Diameter ratio of capillaries of different lengths and diameters to the viscosity was tough to predict. The longer length capillaries would increase the travel of the fluid. The larger diameter capillary would increase the cross-sectional area to which the Carbopol is exposed. The Carbopol has a property of low intermolecular forces which results in less cohesion between the molecules whereas, a significant amount of adhesion is observed i.e. it easily clings to the surface it meets. Hence, it was genuine to predict that increasing the diameter will increase more contact to the fluid and thus reducing the velocity of the flow and thus more resulting in being more viscous. The results obtained showed that the trend of the apparent viscosity reached at a max value for one value of L/D at each pressure and then dropped.

The n values (parameter) from Oswald-de Waele equation were obtained to be 0.39, 0.2701 and 0.2606 for lengths of 3-, 6- and 9-inch capillaries. The plot of shear stress vs velocity gradient of all different lengths did not show the linear relationship. Hence, it can be said that the Carbopol solution is a pseudo plastic non-Newtonian fluid.

The apparent viscosities vs the L/D graph showed the same trend for all the pressures. Thus, it can be concluded that the apparent viscosity decreases as the L/D ratio decreases. The L/D reduction causes the shear stress of a pseudo plastic fluid to decrease and that the fluid of Carbopol solution shows a pseudo plastic behavior.

References

- [1] Non-Newtonian Viscosity Lab Manual, Fall 2017, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.
- [2] McCabe, Warren, et al. *Unit Operations of Chemical Engineering*. 7th ed., McGraw-Hill, 2004
- [3] Bird, R. Byron, et al. *Transport phenomena*. Wiley, 2007
- [4] Towler, Gavin P., and R. K. Sinnott. *Chemical engineering design: principles, practice, and economics of plant and process design*. Elsevier BH, 2013.
- [5] Dumitriu, Severian. *Polymeric biomaterials*. Dekker, 2002.

Appendix I: Data tables & Graphs

Table 1.1: Main Batch Calculations

pH	NAOH Mass (g)	Theoretical Volume (mL)	Theoretical Density (g/mL)	Actual Volume (mL)	Actual Density (g/mL)	total NAOH (mL)
7	16.2	2000	0.0081	2177.50	0.0074	177.5

Table 1.1A: Main Batch NaOH 10 N & 1N supplements

total NAOH (mL)	NaOH 10 n(ml)	NaOH 1N (mL)
177.5	82.5	95

Table 2: Pressure 1 : Fundamental Values at Pressure 1 (30 PSIG (206843 Pa)) and Length 1 (0.0762m)

Trial	Length (m)	Diameter (m)	Radius (m)	L/D (m)	Time (s)	Mass (g)	“Q” Actual Volumetric Flow Rate (m ³ /s)	Actual Log(Q/ $\pi \cdot R^3$)	Log($\Delta PR/2L$)
1	0.0762	0.001778	0.000889	42.85714	30	33.37	0.00015	4.83	3.08
2	0.0762	0.001778	0.000889	42.85714	30	65.76	0.000295	5.13	3.21
3	0.0762	0.001778	0.000889	42.85714	30	109.45	0.00049	5.35	3.21
4	0.0762	0.001016	0.000508	75	30	3.54	1.59E-05	4.59	2.96
5	0.0762	0.001016	0.000508	75	30	1.69	7.57E-06	4.26	2.84
6	0.0762	0.001016	0.000508	75	30	6.19	2.77E-05	4.83	3.06
7	0.0762	0.001575	0.000787	48.3871	30	27.06	0.000121	4.90	3.15
8	0.0762	0.001575	0.000787	48.3871	30	15.86	7.11E-05	4.67	3.03
9	0.0762	0.001575	0.000787	48.3871	30	44.07	0.000197	5.11	3.25

Table 2.1: Pressure 2 : Fundamental Values at Pressure 2 (40 PSIG (275790 Pa)) and Length 2 (0.1524m)

<u>Trial</u>	<u>Length (m)</u>	<u>Diameter (m)</u>	<u>Radius (m)</u>	<u>L/D (m)</u>	<u>Time (s)</u>	<u>Mass (g)</u>	<u>“Q” Actual Volumetric Flow Rate (m³/s)</u>	<u>Actual Log(Q/πR³)</u>	<u>Log(ΔPR/2L)</u>
1	0.1524	0.001778	0.000889	85.71429	30	6.32	2.83E-05	4.11	3.00
2	0.1524	0.001778	0.000889	85.71429	30	14.69	6.58E-05	4.47	3.00
3	0.1524	0.001778	0.000889	85.71429	30	25.65	0.000115	4.72	3.00
4	0.1524	0.013462	0.006731	11.32075	30	1.26	5.65E-06	0.77	3.88
5	0.1524	0.013462	0.006731	11.32075	30	3.04	1.36E-05	1.15	3.88
6	0.1524	0.013462	0.006731	11.32075	30	6.06	2.72E-05	1.45	3.88
7	0.1524	0.001575	0.000787	96.77419	30	3.1	1.39E-05	3.96	2.95
8	0.1524	0.001575	0.000787	96.77419	30	7.47	3.35E-05	4.34	2.95
9	0.1524	0.001575	0.000787	96.77419	30	14.18	6.35E-05	4.62	2.95

Table 2.2: Pressure 3 : Fundamental Values at Pressure 3 (50 PSIG (344738 Pa)) and Length 3 (0.2286)

<u>Trial</u>	<u>Length (m)</u>	<u>Diameter (m)</u>	<u>Radius (m)</u>	<u>L/D (m)</u>	<u>Time (s)</u>	<u>Mass (g)</u>	<u>"Q" Actual Volumetric Flow Rate (m³/s)</u>	<u>Actual Log(Q/πR³)</u>	<u>Log(ΔPR/2L)</u>
1	0.2286	0.001778	0.000889	128.5714	30	2.14	9.59E-06	3.64	2.60
2	0.2286	0.001778	0.000889	128.5714	30	4.53	2.03E-05	3.96	2.73
3	0.2286	0.001778	0.000889	128.5714	30	9.14	4.1E-05	4.27	2.83
4	0.2286	0.013462	0.006731	16.98113	30	0.44	1.97E-06	0.31	3.48
5	0.2286	0.013462	0.006731	16.98113	30	1.06	4.75E-06	0.70	3.61
6	0.2286	0.013462	0.006731	16.98113	30	2.25	1.01E-05	1.02	3.71
7	0.2286	0.001575	0.000787	145.1613	30	0.82	3.67E-06	3.38	2.55
8	0.2286	0.001575	0.000787	145.1613	30	2.44	1.09E-05	3.85	2.68
9	0.2286	0.001575	0.000787	145.1613	30	4.98	2.23E-05	4.16	2.77

Table 3 : Graph Values for m, n & b

<u>Pressure (Pa)</u>	<u>m (Pa*s)</u>	<u>Slope of line "n"</u>	<u>Y-intercept "b"</u>
206843	8.042589	0.39	1.1961
275790	8766.704	0.2701	4.166
344738	3239.135	0.2606	3.728

Table 4: Shear Rates, Apparent Viscosity, Velocity Gradient, and Shear Stress at Pressure 1 : 206843 Pa

<u>Trial</u>	<u>Actual Shear Rate(1/s)</u>	<u>Actual Apparent Viscosity (Pa*s)</u>	<u>Velocity Gradient (1/s)</u>	<u>Shear Stress (Pa)</u>
1	67770.89	0.009	5.90E+15	1.14E+07
2	12835.24	0.025	4.37E+14	4.12E+06
3	4346.11	0.049	8.04E+13	2.13E+06
4	18394.49	0.020	7.68E+14	5.14E+06
5	5.90	2.725	2.63E+09	3.80E+04
6	2.06	5.177	5.07E+08	2.00E+04
7	46356.35	0.011	3.26E+15	9.03E+06
8	9060.82	0.031	2.54E+14	3.33E+06
9	2396.73	0.070	3.17E+13	1.48E+06

Table 4.1: Shear Rates, Apparent Viscosity, Velocity Gradient, and Shear Stress at Pressure 2 : 275790 Pa

<u>Trial</u>	<u>Actual Shear Rate(1/s)</u>	<u>Actual Apparent Viscosity (Pa*s)</u>	<u>Velocity Gradient (1/s)</u>	<u>Shear Stress (Pa)</u>
1	133551.50	1.591	4.83E+03	8.67E+04
2	29833.81	4.751	8.42E+01	2.90E+04
3	9199.94	11.212	3.50E+00	1.23E+04
4	38530.48	3.941	1.68E+02	3.50E+04
5	14.22	1262.531	8.89E-08	1.09E+02
6	4.96	2724.085	5.16E-09	5.06E+01
7	79092.23	2.332	1.17E+03	5.91E+04
8	21833.66	5.966	3.62E+01	2.31E+04
9	7131.75	13.502	1.76E+00	1.02E+04

Table 4.2: Shear Rates, Apparent Viscosity, Velocity Gradient, and Shear Stress at Pressure 3 : 344738 Pa

<u>Trial</u>	<u>Actual Shear Rate(1/s)</u>	<u>Actual Apparent Viscosity (Pa*s)</u>	<u>Velocity Gradient (1/s)</u>	<u>Shear Stress (Pa)</u>
1	222281.20	0.361	2.11E+08	4.78E+05
2	52092.40	1.054	3.43E+06	1.64E+05
3	18562.36	2.260	1.84E+05	7.63E+04
4	67373.91	0.871	7.12E+06	1.98E+05
5	28.35	273.122	1.88E-03	6.31E+02
6	10.53	568.217	1.13E-04	3.03E+02
7	128809.85	0.540	4.48E+07	3.19E+05
8	41445.96	1.248	1.79E+06	1.38E+05
9	14555.78	2.706	9.22E+04	6.37E+04

Table 5: Supplement for Graphs of Actual Apparent Viscosity Vs L/D at Pressure 1: 206843 Pa

<u>L/D (m)</u>	<u>Actual Apparent Viscosity (Pa*s)</u>
11.32	2.73
16.98	5.18
42.86	0.01
48.39	0.01
75.00	0.02
85.71	0.03
96.77	0.03
128.57	0.05
145.16	0.07

Table 5.1: Supplement for Graphs of Actual Apparent Viscosity Vs L/D at Pressure 2: 275790 Pa

<u>L/D (m)</u>	<u>Actual Apparent Viscosity (Pa*s)</u>
11.32	1262.53
16.98	2724.08
42.86	1.59
48.39	2.33
75.00	3.94
85.71	4.75
96.77	5.97
128.57	11.21
145.16	13.50

Table 5.2: Supplement for Graphs of Actual Apparent Viscosity Vs L/D at Pressure 3: 344738 Pa

<u>L/D (m)</u>	<u>Actual Apparent Viscosity (Pa*s)</u>
11.32	273.12
16.98	568.22
42.86	0.36
48.39	0.54
75.00	0.87
85.71	1.05
96.77	1.25
128.57	2.26
145.16	2.71

Table 5.2: Supplement for Graphs of Average Apparent Viscosity Vs Average Shear Rate

<u>Average Velocity Gradient (1/s)</u>	<u>Average Shear Rate</u>
8.46E+12	1.81E+06
1.43E+12	9.05E+05
5.05E+11	6.03E+05

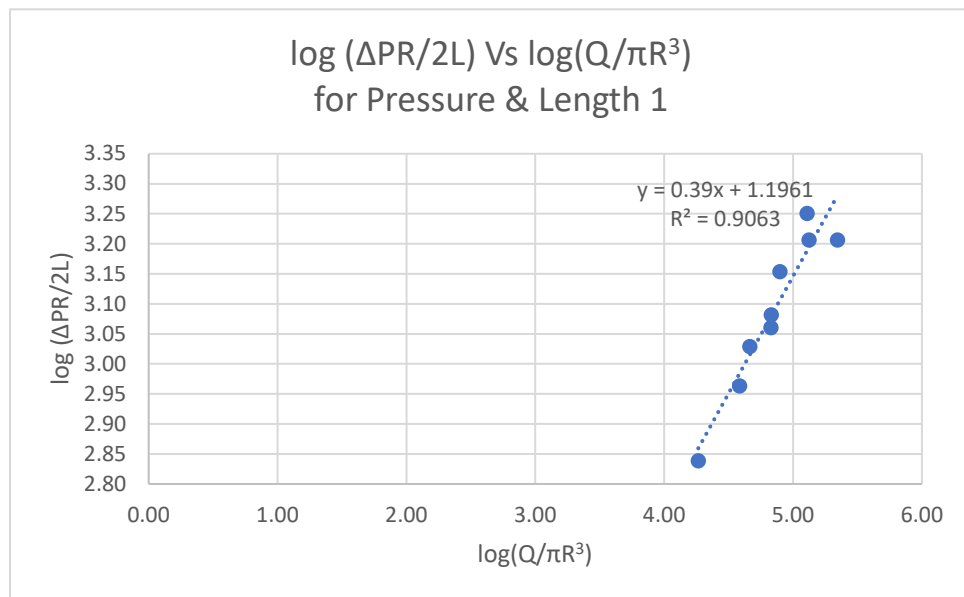


Figure 1: The relation of pressure drop vs flowrate for 3in pipe length and 30 psig

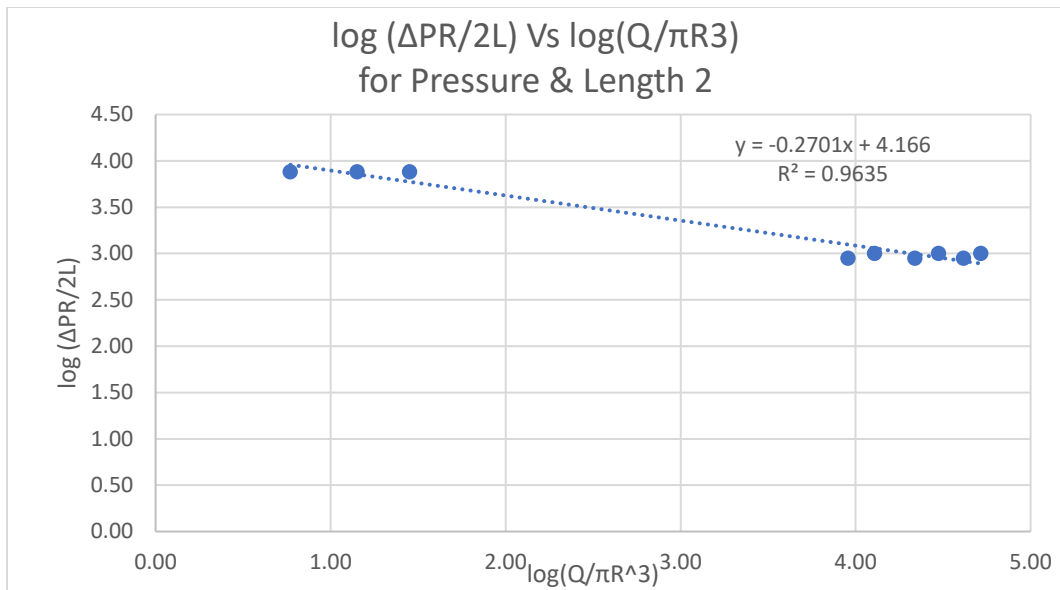


Figure 2: The relation of pressure drop vs flowrate for 6in pipe length and 40 psig

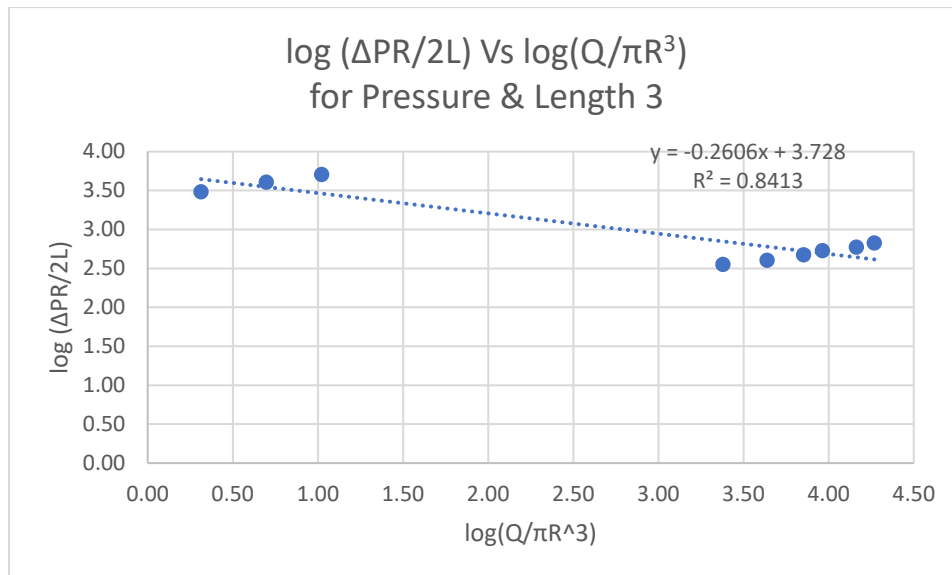


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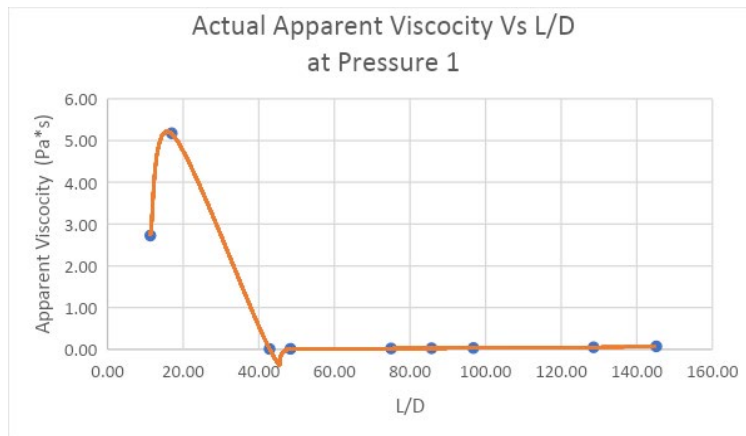


Figure 4: Actual apparent viscosity vs L/D @ 30 psig

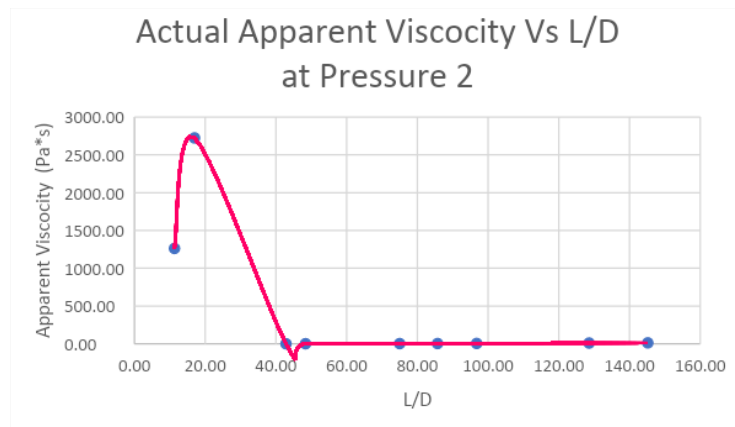


Figure 5: Actual apparent viscosity vs L/D @ 40 psig

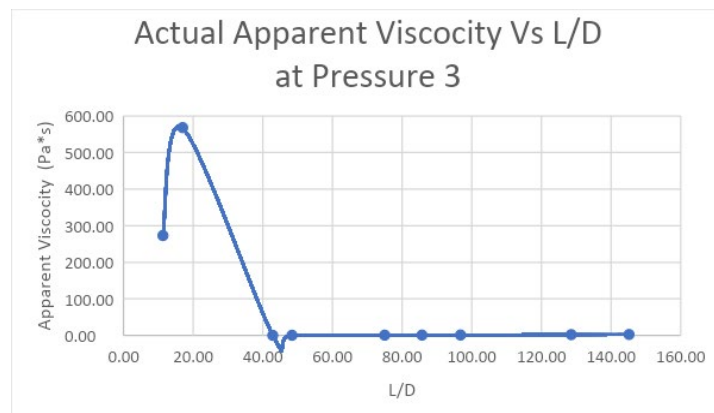


Figure 6: Actual apparent viscosity vs L/D @ 50 psig

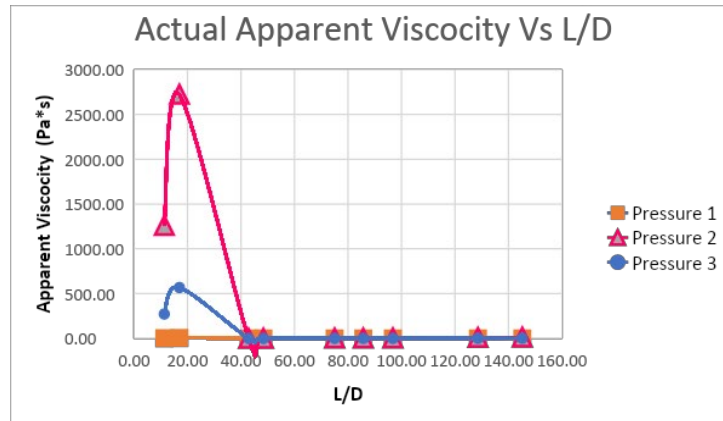


Figure 7: Actual apparent viscosity vs L/D @ combined pressures

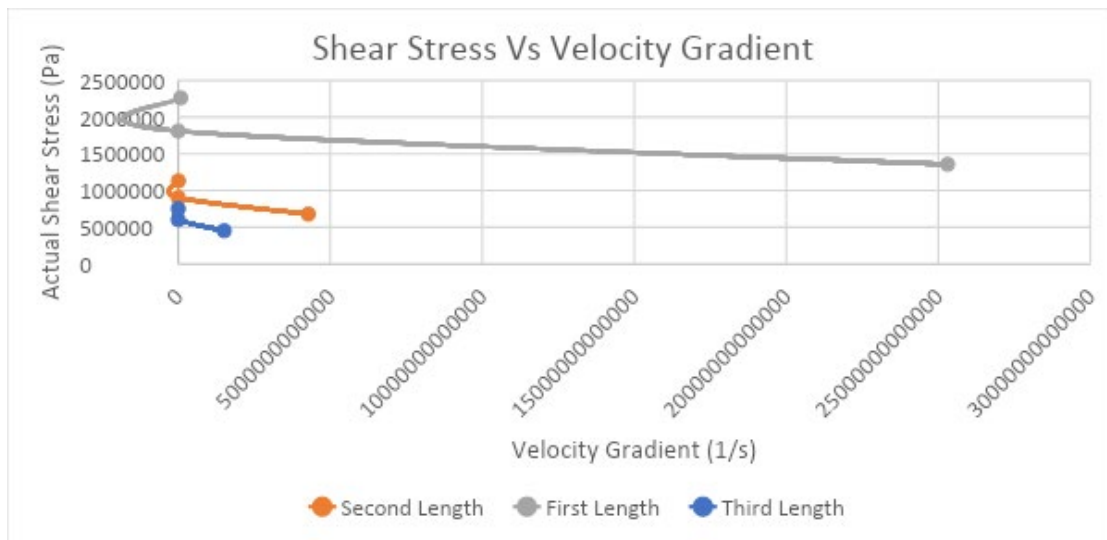


Figure 8: Shear stress versus velocity gradient at 3 in, 6 in and 9 in capillary tube

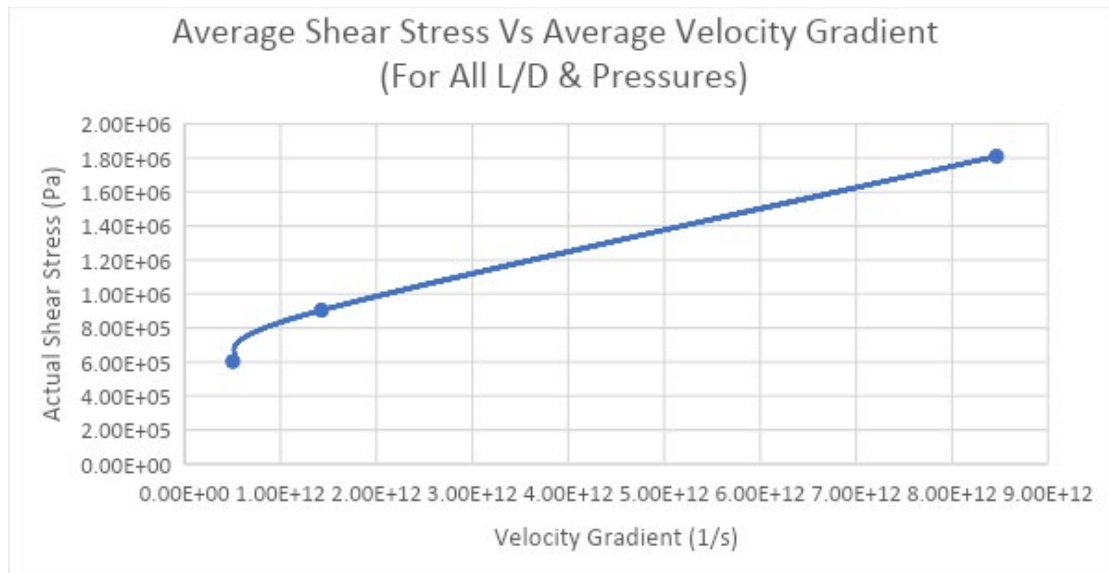


Figure 9: Average shear stress versus average velocity gradient for all L/D and Pressures

Appendix II: Error Analysis

Initial Batch Errors:

When preparing the initial solution of 0.8 wt.% solution of CARBOPOL an error occurred by initially adding the wrong NaOH solution of 10 N and not 1N. A 82.5:95 ratio of 10N to 1N was added to correct the initial error. In turn the weight percent and density was vaguely affected as it became diluted. Yet, noted by tables E1-3 the percent errors are fairly low (~8%) and consistent for the density, volume and rates readings. In turn it may be suggested that the errors do not significantly effect the accuracy of results. The ratio was deemed suitable once the pH of 7 was reached.

Experimental Errors:

Time of 30 seconds was deemed accurate but may have varied from trial to trail vaguely, since collections were done by lab partners and not a systematic machine approach. Material and supply errors may have occurred as mentioned in the table below. Lastly, the standard deviations are fairly low, and fair based on the range of data.

Table E: Initial Errors:

<u>Equipment</u>	<u>Uncertainty</u>	<u>Comment</u>
Graduated Cylinder (100ml)	± 0.5 mL	The measurements were eyeballed from the lower meniscus between 0.5 mL lines
Scale	$\pm .1$ g	Scale is calibrated to 0.1 g
Stop Watch	± 0.01 s	The stop watch may be off by 0.01 s
pH Strips	Color shade	Strips are accurate to one color shade
Pressure reading	± 1 PSIG	Accurate to 1 PSIG based on eyeball readings

Table E1: Errors for Pressure 1:

<u>Trials</u>	<u>Length (m)</u>	<u>Diameter (m)</u>	<u>Theoretical Density of gel (g/mL)</u>	<u>Actual Density (g/mL)</u>	<u>% Error</u>	<u>Actual Volume (mL)</u>	<u>Theoretical Volume of gel (mL)</u>	<u>% Error</u>
1	0.0762	0.001778	0.0081	0.0074	8.151	4,485.38	4,119.75	8.87
2	0.0762	0.001778	0.0081	0.0074	8.151	849.49	780.25	8.87
3	0.0762	0.001778	0.0081	0.0074	8.151	287.65	264.20	8.87
4	0.0762	0.001016	0.0081	0.0074	8.151	227.16	208.64	8.88
5	0.0762	0.001016	0.0081	0.0074	8.151	169.36	155.56	8.87
6	0.0762	0.001016	0.0081	0.0074	8.151	59.14	54.32	8.88
7	0.0762	0.0015748	0.0081	0.0074	8.151	2,131.80	1,958.02	8.88
8	0.0762	0.0015748	0.0081	0.0074	8.151	416.68	382.72	8.88
9	0.0762	0.0015748	0.0081	0.0074	8.151	110.22	101.23	8.88
		<u>Average</u>	0.0081	0.0074	8.1515	970.76	891.63	8.8750
		<u>Standard Deviation</u>	0.0000	0.0000	0.0000	1,468.66	1,348.94	0.0000
<u>Trials</u>	<u>Length (m)</u>	<u>Diameter (m)</u>	<u>Theoretical Volumetric Flow Rate (mL/s)</u>	<u>Actual Volumetric Flow Rate (mL/s)</u>	<u>% Error</u>	<u>Theoretical Volumetric Flow Rate (m³/s)</u>	<u>Actual Volumetric Flow Rate (m³/s)</u>	<u>% Error</u>
1	0.0762	0.001778	137.33	149.51	8.87	0.00	0.00	8.87
2	0.0762	0.001778	26.01	28.32	8.87	0.00	0.00	8.88
3	0.0762	0.001778	8.81	9.59	8.87	0.00	0.00	8.87
4	0.0762	0.001016	6.95	7.57	8.88	0.00	0.00	8.88
5	0.0762	0.001016	5.19	5.65	8.87	0.00	0.00	8.87

6	0.076 2	0.001016	1.81	1.97	8.88	0.00	0.00	8.88
7	0.076 2	0.001574 8	65.27	71.06	8.87	0.00	0.00	8.88
8	0.076 2	0.001574 8	12.76	13.89	8.87	0.00	0.00	8.88
9	0.076 2	0.001574 8	3.37	3.67	8.88	0.00	0.00	8.88
		<u>Average</u>	29.72	32.36	8.8750	0.00	0.00	8.875 0
		<u>Standard Deviation</u>	44.96	48.96	0.0000	0.00	0.00	0.000 0

Table E2: Errors for Pressure 2:

<u>Trials</u>	<u>Length</u> <u>(m)</u>	<u>Diameter</u> <u>(m)</u>	<u>Theoretical</u> <u>Density</u> <u>of gel</u> <u>(g/mL)</u>	<u>Actual</u> <u>Density</u> <u>(g/mL)</u>	<u>% Error</u>	<u>Actual</u> <u>Volume</u> <u>(mL)</u>	<u>Theoretical</u> <u>Volume</u> <u>of gel</u> <u>(mL)</u>	<u>% Error</u>
1	0.152 4	0.001778	0.0081	0.0074	8.151	8,839.04	8,118.52	8.88
2	0.152 4	0.001778	0.0081	0.0074	8.151	1,974.54	1,813.58	8.87
3	0.152 4	0.001778	0.0081	0.0074	8.151	608.89	559.26	8.87
4	0.152 4	0.001016	0.0081	0.0074	8.151	475.82	437.04	8.88
5	0.152 4	0.001016	0.0081	0.0074	8.151	408.62	375.31	8.88
6	0.152 4	0.001016	0.0081	0.0074	8.151	142.48	130.86	8.88
7	0.152 4	0.001574 8	0.0081	0.0074	8.151	3,637.23	3,340.74	8.88
8	0.152 4	0.001574 8	0.0081	0.0074	8.151	1,004.07	922.22	8.87
9	0.152 4	0.001574 8	0.0081	0.0074	8.151	327.97	301.23	8.88
	0.152 4	<u>Average</u>	0.0081	0.0074	8.1510	1,935.41	1,777.64	8.875 0

	0.1524	<u>Standard Deviation</u>	0.0000	0.0000	0.0000	2,816.13	2,586.57	0.0000
<u>Trial</u>		<u>Diameter (m)</u>	<u>Theoretical Volumetric Flow Rate (mL/s)</u>	<u>Actual Volumetric Flow Rate (mL/s)</u>	<u>% Error</u>	<u>Theoretical Volumetric Flow Rate (m³/s)</u>	<u>Actual Volumetric Flow Rate (m³/s)</u>	<u>% Error</u>
1	0.1524	0.001778	270.62	294.63	8.88	0.00	0.00	8.88
2	0.1524	0.001778	60.45	65.82	8.87	0.00	0.00	8.87
3	0.1524	0.001778	18.64	20.30	8.87	0.00	0.00	8.88
4	0.1524	0.001016	14.57	15.86	8.88	0.00	0.00	8.87
5	0.1524	0.001016	12.51	13.62	8.88	0.00	0.00	8.88
6	0.1524	0.001016	4.36	4.75	8.88	0.00	0.00	8.88
7	0.1524	0.0015748	111.36	121.24	8.88	0.00	0.00	8.88
8	0.1524	0.0015748	30.74	33.47	8.88	0.00	0.00	8.87
9	0.1524	0.0015748	10.04	10.93	8.87	0.00	0.00	8.87
		<u>Average</u>	59.25	64.51	8.8750	0.00	0.00	8.8750
		<u>Standard Deviation</u>	86.22	93.87	0.0000	0.00	0.00	0.0000

Table E3: Errors for Pressure 3:

<u>Trial</u>	<u>Length (m)</u>	<u>Diameter (m)</u>	<u>Theoretical Density of gel (g/mL)</u>	<u>Actual Density (g/mL)</u>	<u>% Error</u>	<u>Actual Volume (mL)</u>	<u>Theoretical Volume of gel (mL)</u>	<u>% Error</u>
1	0.2286	0.001778	0.0081	0.0074	8.151	14,711.57	13,512.35	8.88
2	0.2286	0.001778	0.0081	0.0074	8.151	3,447.71	3,166.67	8.88

3	0.228 6	0.001778	0.0081	0.0074	8.151	1,228.54	1,128.40	8.88
4	0.228 6	0.001016	0.0081	0.0074	8.151	832.02	764.20	8.88
5	0.228 6	0.001016	0.0081	0.0074	8.151	814.55	748.15	8.88
6	0.228 6	0.001016	0.0081	0.0074	8.151	302.43	277.78	8.88
7	0.228 6	0.001574 8	0.0081	0.0074	8.151	5,923.61	5,440.74	8.88
8	0.228 6	0.001574 8	0.0081	0.0074	8.151	1,905.98	1,750.62	8.88
9	0.228 6	0.001574 8	0.0081	0.0074	8.151	669.38	614.81	8.87
		<u>Average</u>	0.0081	0.0074	8.151	3,315.09	3,044.86	8.875 0
		<u>Standard Deviation</u>	0.0000	0.0000	0.0000	4,630.54	4,253.08	0.000 0
<u>Trials</u>	<u>Length (m)</u>	<u>Diameter (m)</u>	<u>Theoretical Volumetric Flow Rate (mL/s)</u>	<u>Actual Volumetric Flow Rate (mL/s)</u>	<u>% Error</u>	<u>Theoretical Volumetric Flow Rate (m³/s)</u>	<u>Actual Volumetric Flow Rate (m³/s)</u>	<u>% Error</u>
1	0.228 6	0.001778	450.41	490.39	8.87	0.00	0.00	8.87
2	0.228 6	0.001778	105.56	114.92	8.88	0.00	0.00	8.87
3	0.228 6	0.001778	37.61	40.95	8.88	0.00	0.00	8.88
4	0.228 6	0.001016	25.47	27.73	8.88	0.00	0.00	8.87
5	0.228 6	0.001016	24.94	27.15	8.88	0.00	0.00	8.88
6	0.228 6	0.001016	9.26	10.08	8.88	0.00	0.00	8.88
7	0.228 6	0.001574 8	181.36	197.45	8.88	0.00	0.00	8.87
8	0.228 6	0.001574 8	58.35	63.53	8.87	0.00	0.00	8.87
9	0.228 6	0.001574 8	20.49	22.31	8.87	0.00	0.00	8.87
		<u>Average</u>	101.50	110.50	8.8750	0.00	0.00	8.875 0

		<u>Standard</u> <u>Deviation</u> <u>n</u>	141.77	154.35	0.0000	0.00	0.00	0.000 0
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Appendix III: Sample Calculations

1. Density Calculation

The following calculations are done using the values for 3in length pipe at 30 psig.

$$\text{Density of the Carbopol solution} = \frac{\text{Actual Mass of carbopol}}{\text{Volume of carbopol}} = \frac{16.2}{2177.50} = 0.0074 \frac{g}{mL}$$

From the experimental data we have,

Actual mass of the Carbopol = 33.37 g

Density of the Carbopol = 0.01 g/mL

Time duration = 30 s

$$\text{Volume of the carbopol} = \frac{\text{Actual mass of carbopol}}{\text{density of the carbopol}} = 4485.38 \text{ mL}$$

2. Volumetric Flow rate (Q) Calculation

$$Q = \frac{\text{volume of the carbopol}}{\text{time}} = \frac{4485.37}{30} = 149.51 \frac{mL}{s}$$

Volumetric flow rate can be calculated in SI unit:

$$Q = 149.51 \frac{mL}{s} * \frac{1 \text{ m}^3}{10^6 \text{ mL}} = 0.00015 \frac{\text{m}^3}{s}$$

3. Oswald – de – Waele Parameters Calculation (m and n)

The m and n parameters were measured after plotting the ln-ln graph of $\log(dP R/2L)$ versus $\log(Q/\pi R^3)$. Using the linear fitting, the required values can be determined. Parameter 'n' is the slope of the line. For the 3, 6 and 9 in capillaries, the n values were found to be 0.39, 0.2701, and 0.2606 respectively. Also, the m value can be calculated using the y-intercept of the trendline equation by using,

$$m = \frac{10^b}{\left(\frac{3n+1}{n}\right)^n} = \frac{10^{1.1961}}{\left(\frac{3(0.39)+1}{0.39}\right)^{0.39}} = 8.0425 \text{ Pa} \cdot \text{s}$$

Where b is the y-intercept of the trendline.

4. Apparent Viscosity (μ)

$n = 0.39$ and $m = 8.042589$ were obtained above.

$$Area = \frac{\pi}{4} D^2 = \frac{\pi}{4} (0.001778)^2 = 2.48 \times 10^{-6} \text{ m}^2$$

The shear rate can also be determined using,

$$\text{Shear Rate } (\dot{\gamma}) = \frac{dV_x}{dy} = \frac{\text{Volumetric Flow rate}}{\text{Area} * \text{Radius}} = \frac{0.00015}{2.48 \times 10^{-6} * 0.001} = 67770.89 \text{ s}^{-1}$$

Also, the apparent viscosity is given as,

$$\text{Apparent viscosity} = \eta(\dot{\gamma}) = m \dot{\gamma}^{n-1}$$

$$\text{Apparent viscosity} = 8.04259 * (67770.89)^{0.39-1} = 0.009 \text{ Pa} \cdot \text{s}$$

5. Shear Stress Calculation

The following formula can be utilized to calculate the shear stress,

$$\tau_{rz} = \pm m \left| \frac{dv_z}{dr} \right|^n$$

The velocity gradient can be calculated as,

$$\frac{dv_z}{dr} = \left(\frac{\Delta P}{m * 2L} \right)^{\frac{1}{n}} = 5.90 * 10^{15} \text{ s}^{-1}$$

Utilizing above equation, shear stress was found to be,

$$\tau_{rz} = 1.14 * 10^7 Pa$$

Appendix IV: Team Member Contributions

NAME: Daniela Estarita

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Myself and Nisha stayed to prepare the solution on prelab day and clean up. Worked as a team on lab day
Pre-Lab Editing	.3	Edited my portion + looked over others
Experimental Objective (Prelab)	n/a	Split portions as a group
Apparatus (prelab)	0.15	Worked on
Materials & Supplies (prelab)	0.45	Worked on
Procedure (prelab)	n/a	Split portions as a group
Project Safety Evaluation (prelab)	0.45	Worked on
Final Lab Editing	2.5	Read all portions/edited my portions, included team contributions, cover page
Executive Summary (final lab)	n/a	Split portions as a group
Introduction (Final lab)	n/a	Split portions as a group
Theory (Final Lab)	n/a	Split portions as a group
Results (final lab)	1	Prepared data for solutions along with errors/graphs to reference for results
Discussion (final lab)	n/a	Split portions as a group
References (final and/or prelab)	n/a	Split portions as a group
Data Tabulation/Graphs (final)	9	Did calculations
Error Analysis (final lab)	3	Calculated errors & wrote about them
Sample Calculations (final Lab)	0.2	Expanded upon lab manuals
Power Point Presentation	0.25	Worked on safety & materials/supply slide
Total	25.3	

NAME: Huda Usman

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Worked as a team on both days. Had to leave a little early in the prelab day for a meeting with an advisor
Pre-Lab Editing	3	Finished my part and looked over others
Experimental Objective (Prelab)	0.30	Worked on
Apparatus (prelab)	1	Worked on
Materials & Supplies (prelab)	0.30	Worked with another team member
Procedure (prelab)	0.30	Worked with another team member
Project Safety Evaluation (prelab)	0.25	Looked over and helped edit another team member
Final Lab Editing	5	Wrote introduction, did the sample calculations, looked over results and discussion and made few changes
Executive Summary (final lab)	0	Assigned to another member
Introduction (Final lab)	2	Worked on
Theory (Final Lab)	0.15	Helped edited the other teammate
Results (final lab)	n/a	Assigned to another member
Discussion (final lab)	n/a	Assigned to another member
References (final and/or prelab)	0.15	Used citations and check the numbering
Data Tabulation/Graphs (final)	0.15	Looked over the values while doing sample calculations and made necessary changes
Error Analysis (final lab)	n/a	Assigned to another member
Sample Calculations (final Lab)	2	Worked on
Power Point Presentation	1	Worked on
Total	22.60	

NAME: Nisha Patel

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Daniela and I stayed to prepare the solution on prelab day and clean up. Worked as a team on lab day
Pre-Lab Editing	.5	Edited my portion + looked over others
Experimental Objective (Prelab)	n/a	Split portions as a group
Apparatus (prelab)	0.45	Worked on
Materials & Supplies (prelab)	0.45	Worked on
Procedure (prelab)	n/a	Split portions as a group
Project Safety Evaluation (prelab)	0.20	Worked on
Final Lab Editing	2.5	Read all portions/edited my portions, included team contributions, cover page
Executive Summary (final lab)	n/a	Split portions as a group
Introduction (Final lab)	n/a	Split portions as a group
Theory (Final Lab)	4	Split portions as a group
Results (final lab)	n/a	Split portions as a group
Discussion (final lab)	5	Worked on discussion based on the results obtained
References (final and/or prelab)	0.30	Worked on it
Data Tabulation/Graphs (final)	n/a	Split portions as a group
Error Analysis (final lab)	n/a	Split portions as a group
Sample Calculations (final Lab)	n/a	Split portions as a group
Power Point Presentation	0.5	Worked on procedure & materials/supply slide
Total	21.9	

NAME: Mike Topalis

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	
Pre-Lab Editing	0.30	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0	
Executive Summary (final lab)	2	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	2	
Discussion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	
Total	8.30	